

A Beginner's Guide to Wireshark, the Top-rated Packet Sniffer

Wireshark is the world's foremost network protocol analyser. It lets users see what is happening in the network at a microscopic level. It is used for network troubleshooting, analysis, software and communications protocol development and education.

Wireshark is a valuable tool for network administrators and enthusiasts, which is used to capture the packets in real-time on any operating system. This enables admins to understand the functions of various protocols, to troubleshoot live network problems, and monitor the network for intrusion detection. It can also be used for a software defined network (SDN).

Wireshark is application software that helps you to visualise the traffic on the network interface controller. Data can be read from the captured packets using a graphical user interface (GUI). It can capture packets in promiscuous mode (all the packets in a LAN), and dissect more than a thousand protocols.

Wireshark is free and open source under the terms of the GNU (General Public License). It is cross-platform and helps us understand the structure of various network protocols. It uses the Pcap (packet capture library) to capture packets from different network technologies; raw USB traffic and compatible encoded VoIP calls can also be captured and played.

Wireless tools are used to capture wireless network traffic. Wireshark captures packets from network simulation tools like ns, OPNET, NeTSim, etc. Users can customise this packet sniffer based on their needs. It gives statistical results of the packets captured, as well as graphs such as the I/O graph, round-trip time graph and flow graph. It gives us expert information to understand the state of the packet using a particular protocol category like chat, note, warning, error, etc. This software is widely used in industries and academic institutions.

Wireshark loads the Pcap driver, which uses NDIS (Network Driver Interface Specification) to read packets directly from a network interface card.

In this article, we will figure out how to install and use Wireshark for the Windows operating system.

The latest version of Wireshark can be downloaded from <https://www.wireshark.org>. There are also options available for other operating systems.

How to install Wireshark

The downloaded executable file should be run to install Wireshark.

You can capture the packets with the default options by selecting the appropriate interface. Here, we have selected the Local Area Connection interface (highlighted). Click the shark icon in the top left corner to start capturing.

If you want to modify the default options, you can do so before you choose the interface. There are options available to perform various name resolutions (Figure 5).

Wireshark consists of three windows – the packet list pane, the packet details pane and the packet bytes pane, as shown in Figure 6. The packet list pane displays the live frames captured in columns. The serial number of the frame captured, the time, the IPv4 address of the source machine, the destination machine, the protocol involved, the length of the frame and the information column that describes the frame captured are given in the consecutive columns. Users can customise these columns based on their interests.